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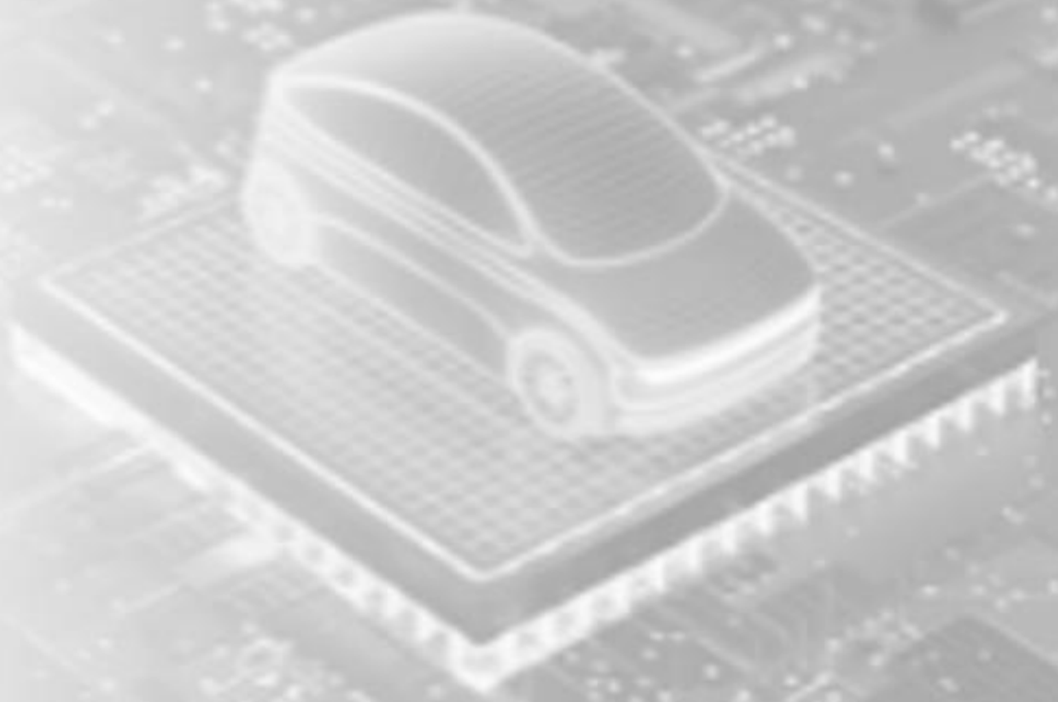
SEMICONDUCTORS

Automotive
Fabless
Company

LLMs Development Kickoff (Proposal Plan)



Created date: 2025-08-04

V.Nhat.

Agenda

1. Introduction
2. State of Play
3. Proof of Concept Plan

Introduction

Infotainment

Why it? Why us? Why now?

In-Vehicle Infotainment (IVI)

- **Infotainment as a Smart Companion**
  - Voice, vision, and AI assistants
  - Natural, safe, and proactive interaction
- **BMW + Alexa LLM:** BMW Will Enhance the iDrive 9
- **BMW + Alibaba's Qwen (China):** BMW's Neue Klasse Will Run On Alibaba's AI To Win Back China | Carscoops
- **Infotainment = Luxury automotive**

In-Vehicle Infotainment (IVI)

- **BMW + Amazon's Alexa LLM**: BMW Will Enhance the iDrive 9
    - Integrated, but *cloud-dependent*
- **BMW + Alibaba's Qwen (China):** BMW's Neue Klasse Will Run On Alibaba's AI To Win Back China | Carscoops
    - Localized, but *fragmented*
- **BMW + Qualcomm:**
    - Strong chips, but highly *dominate* given their monopoly

- **BOS Semiconductors:**
    - *Offline*, *unified*, and *controllable* LLM stack with more *affordable* chips
    - Sovereignty – BMW-branded assistant & Negotiation Leverage

State of Play

LLMs

Hardware

Optimization

State of Play

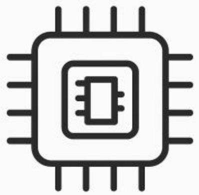
Current Industry Moves

- Amazon Alexa integrated into BMW iDrive (US/EU).
- Alibaba Qwen powering iDrive X in China.
- Qualcomm supplying SoCs for infotainment baseline.

Next-gen IVI

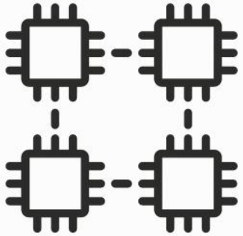
- Unified, on-device runtime across global markets.
- Scalable from 1 device -> multi-device (premium).
- One BMW-branded assistant.

BHRC PoC = Capability Showcase -- *First proof BMW can own in-vehicle AI, not outsource it* --



Model fit
& Scaling

Latency/
Offline reliability



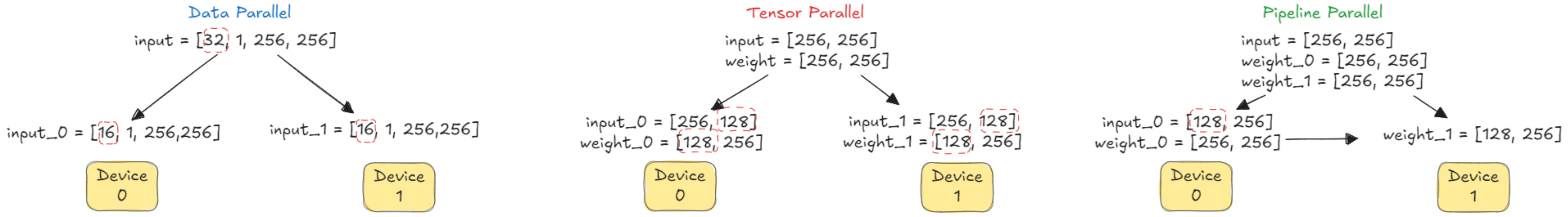
Device-mesh
Orchestration

Misc.

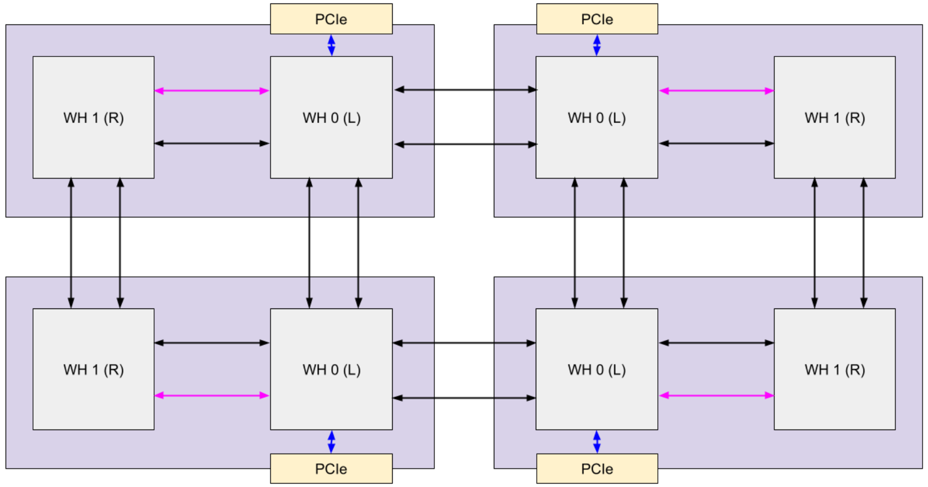
Model Exploration

- **Candidates:** Qwen, LLaMA, and DeepSeek
- **Evaluation criteria:** Licensing, size/scaling, multilingual support, and implementation, etc.
- **Output:** Shortlist of 1-2 models for PoC.

Multi-device Orchestration



- **Goal:** Multi-device (MeshDevice) Understanding & Experiments & Documentation
  - Topology; Partitioning strategies; Key challenges; Communication between devices
- **Output:** Working multi-device demo
- **References:**
  - Modeling Multi-Device in Tenstorrent's Compiler
  - tenstorrent/tt-metal | DeepWiki
  - tt-metal/tech_reports/Programming_Mesh_of_Devices/Programmi md at main · tenstorrent/tt-metal



Miscellaneous*

- **Goal:** Improve efficiency, fit, and reliability of the LLM stack.
    - **Graph Optimization**: operator fusion, pruning, activation re-computation.
    - **Quantization**
    - **KV-Cache** management: memory footprint reduction (compression, streaming)
- **Deliverables**: Streamline model with optimized model graph & quantized checkpoints

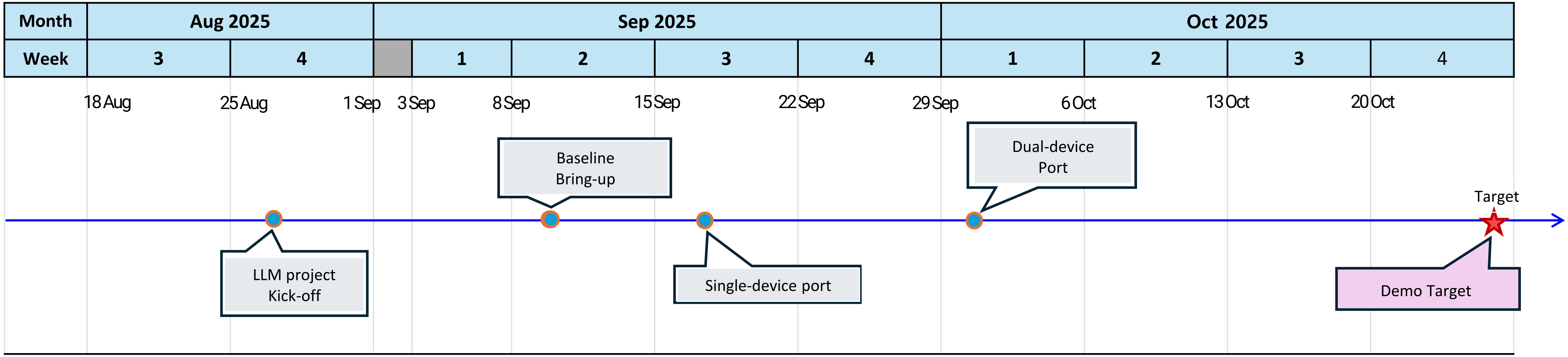
Proof-of-Concept Plan

Goal

Setup

KPIs

PoC Plan



BHRC Capability Showcase:
- Show **multi-device LLM inference** on **2x Wormhole** devices.
- Demonstrate scalability.
- Deliver a **shareable video demo** within ~7 weeks

Success metrics: TBD
- **Latency:** TBD
- **Throughput:** TBD
- **Memory fit:** TBD
- **Scaling:** TBD
- **Functionality:** TBD

PoC Plan

No.	Phase	Due date	Details	Output
1	Kick-off	29-Aug-25	Model Specs study (Qwen)	- 1 final model for PoC - Comparative survey, Model Architecture, Dataset - Performance benchmark (metrics)
			Model Specs study (LLaMA)	
			Model Specs study (DeepSeek)	
		5-Sep-25	TTNN Mesh Device study	Development guideline "Model on MeshDevice"
2	Baseline Bring-up	12-Sep-25	First setup	- Project structure - Development resource allocation across team
			GPU PyTorch implementation	- Updated model specs - Additional constraints and requirements - TTNN Implementation plan (e.g. Gantt Chart)
			CPU PyTorch implementation	
3	*Single-device port*	*26-Sep-25*	*TBD*	*Functional-Enabled Implementation*
4	*Dual-device port*	*10-Oct-25*	*TBD*	*A shareable PoC demo video*

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Thank you